

Phase 4 Report

1. Project Summary

Our project involves investigating how movie discussions might relate to commercial success, and our three questions are as follows:

1. How does Reddit recommendation frequency relate to the proportion of a movie's box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?
2. How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?
3. How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre?

By analyzing box office numbers and other movie specifications in relation to Reddit movie discussions, which itself does not have specific rating-based systems for recommendations, we aim to build a better understanding of what general movie audiences value.

2. MongoDB Schema Design

Collection Name: movies

Top-level fields:

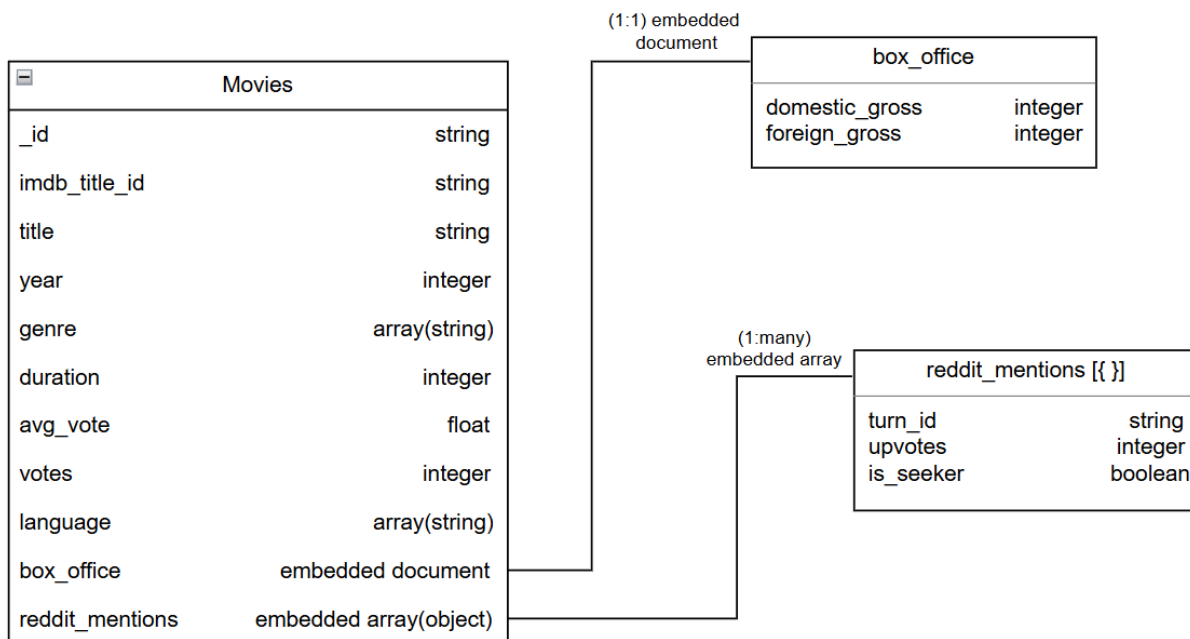
_id (string)
imdb_title_id (string)
title (string)
year (integer)
genre (array of strings)
duration (integer)
avg_vote (float)
votes (integer)
language (array of strings)

Embedded Document: box_office

domestic_gross (number)
foreign_gross (number)
total_gross (number)
- Each movie has exactly one box office record (1:1 relationship)

Embedded Array: reddit_mentions

- Each element has: turn_id (integer), upvotes (integer), is_seeker (boolean)



JSON Documentation:

```
{
  "_id": string,
  "imdb_title_id": string
  "title": string,
  "year": integer,
  "genre":[
    {string},
    {string},
    {string}
  ],
  "duration": integer,
  "avg_vote": float,
  "votes": integer,
  "language":[
    {string},
    {string},
    {string}
  ],
  "box_office":{
    "domestic_gross": integer,
    "foreign_gross": integer
  }
  "reddit_mentions":[
    {
      "turn_id": string,
      "upvotes": integer,
      "is_seeker": boolean
    },
    {
      "turn_id": string,
      "upvotes": integer,
      "is_seeker": boolean
    },
    {
      "turn_id": string,
      "upvotes": integer,
      "is_seeker": boolean
    }
  ],
}
```

3. Data Loading Code

Please refer to “mongoDB_data_loader.ipynb”

4.

RQ1: How does Reddit recommendation frequency relate to the proportion of a movie's box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?

Analysis Query (Cell 2, Lines 2 - 49)

The MongoDB query pipeline filters the movie documents to keep only movies whose genres include horror, action, or comedy, and have non-null domestic and foreign box office values. The pipeline then uses \$project to keep the fields needed for the analysis and to create derived variables. In particular, num_recommendations is calculated as the number of elements in the embedded reddit_mentions array using \$size, and total_revenue is computed by summing the domestic gross and foreign gross. Domestic share and foreign share are calculated by dividing each market's revenue by total revenue. The \$cond operator prevents division by zero by returning None when the total revenue is 0.

The MongoDB query answers the same research question as the SQL query, but it works directly on the nested document structure of the movies collection. In SQL, we needed to join imdb_movies and bom_gross, along with an aggregated subquery of reddit_mentions, to combine genre, box office, and Reddit recommendations data. In MongoDB, an aggregation pipeline was used to join box_office.domestic_gross and num_recommendations directly as the related attributes were already stored together in each movie document. No new joins were needed.

Initially, the MongoDB query returned 34 fewer rows than the SQL query. After comparing the outputs, we found that the SQL query contained duplicate movies. This happened because while writing the SQL query, we initially assumed that combining the title and year would uniquely identify each movie in the SQL table. However, this assumption did not hold, resulting in duplicate (title, year) pairs. In contrast, the MongoDB collection stores one document per imdb_title_id, so each movie appears only once. After identifying this issue, we used DISTINCT in the SQL query to remove the duplicates, and the SQL and MongoDB results matched.

Visualization Query (Cell 5, Lines 2 - 107)

The MongoDB aggregation pipeline creates a dataframe used for visualization by first filtering the movies collection to include only horror, action, and comedy films with non-null domestic and foreign box office values. It uses \$project to select key variables, and the pipeline calculates domestic and international shares using \$cond to prevent zero-division errors. The pipeline groups movies into recommendation-frequency ranges (eg: 0-24, 25-49, etc.) using \$switch.

Since genre is stored as an array, \$unwind is used to separate each genre into its own row, and another filter is used to retain the relevant genres. The data is shaped into long format by combining domestic and international shares into a single market_data field, which is unwound to create one row per market. Finally, the data is grouped by genre, recommendation group, and market to compute the average revenue share and number of groups in each category.

RQ2: How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?

Analysis Query (Cell 7 Lines 2-27)

The MongoDB query first ensures that any observation containing null values for domestic gross or foreign gross is dropped. Additionally, the query filters the documents to keep any movies containing 1 or more Reddit mentions. Since the original SQL query for this question used an inner join between the Reddit and IMDb datasets, filtering out movies with 0 Reddit mentions in this query replicates that mechanism. Including the movies with no Reddit mentions could have provided valuable information for this analysis, and so excluding them is a limitation of this study.

The rest of the MongoDB pipeline is used to calculate the needed values to answer the research question. Using \$project, the values needed in the query's output were calculated and specified: total_gross, avg_vote, and reddit_discussion_count. Total_gross was calculated by using the \$add function to sum each movie's domestic gross and foreign gross, while reddit_discussion_count was calculated by using the \$size function on each movie's array of its reddit mentions to find the total number of reddit mentions per movie. Lastly, avg_vote was left unchanged. This pipeline was then fed into MongoDB's aggregate function to produce results that mirror the SQL query results from phase 3.

The MongoDB query answers the research question in the same way as the SQL query of Phase 3, as they result in the same data. However, the queries took different routes to get to the same place. Where the SQL query needed to use multiple joins to use data across all 3 datasets, the MongoDB data loading code handled this. Due to this, the query could focus on the calculations rather than the joins. The data loading code also grouped all the data by movie; thus, the GROUP BY function in the SQL query was also bypassed in the MongoDB query, allowing calculations to be done right away. Due to this aggregation, the reddit_discussion_count was calculated in a different way, simply by counting the length of each movie's "reddit_mentions" array. Overall, the query for this research question is much easier to read and execute in MongoDB in comparison to SQL due to the heavy lifting from the MongoDB data loading code.

RQ3: How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre (horror, action or comedy)?

Analysis Query (Cell 11, Lines 9-143)

The MongoDB query for this question first uses the operator `$unwind` to separate the Reddit mentions associated with a given movie, so each Reddit mention/post is its own separate document, with a given number of associated upvotes. The query also uses `$unwind` to separate the list of genres so each genre is its own separate document. Following, the query uses the `$match` operator along with the `$in` operator to filter for movies that are in either the horror, action, or comedy genres. The operator `$addFields`, along with the operators `$cond`, `$and`, `$gte`, and `$lte` or `$lt` are used multiple times to bin all the movies by their durations (in minutes), creating a new categorical variable. Next, the `$group` operator groups the data by genre and binned duration, computing a new variable named “avg_votes” as the average number of upvotes on Reddit mentions within each genre-duration bin group. Finally, the query uses the `$project` operator to deselect the document id (“_id”), and select the columns relevant to the research question: “avg_upvotes”, “genre”, and “duration_bin”. Due to the use of the `$unwind` operator on `reddit_mentions` at the beginning of the pipeline, any movies with an empty `reddit_mentions` array are excluded from the results.

Upon comparison of the computed average upvote values, I realized that some groups had different numbers. This is likely due to how the SQL and MongoDB queries filter the movies for the genres of interest. Given a list of genres, the SQL query would classify a movie as belonging to the first relevant genre in its genre list. The genres are thus mutually exclusive, as a movie can only belong to one genre given the SQL query structure. In contrast, the MongoDB query uses the `$unwind` operator on the genre array, so each genre associated with a movie becomes its own document. As a result, the genre categories in MongoDB are not mutually exclusive—if a movie is tagged with more than one relevant genre, that same movie and its Reddit mentions can contribute to multiple genre groups when calculating average number of upvotes. This explains why some average upvote values resulting from the MongoDB query differ slightly from the SQL output (results compared in cell 14).

The MongoDB query answers the research question slightly better than the SQL query due to how it approaches filtering movie genres—in MongoDB, movies that belong to multiple genres are included in each relevant category, ensuring that all associated Reddit upvotes are captured. In contrast, the SQL query assigns a movie to only one genre during filtering, which can exclude some relevant Reddit activity. For example, a movie categorized as both action and comedy would contribute its Reddit mention upvotes to both genres in MongoDB, but only one genre in

SQL. As a result, MongoDB averages reflect a more comprehensive representation of Reddit engagement, while SQL averages are based on a more restricted subset of data.

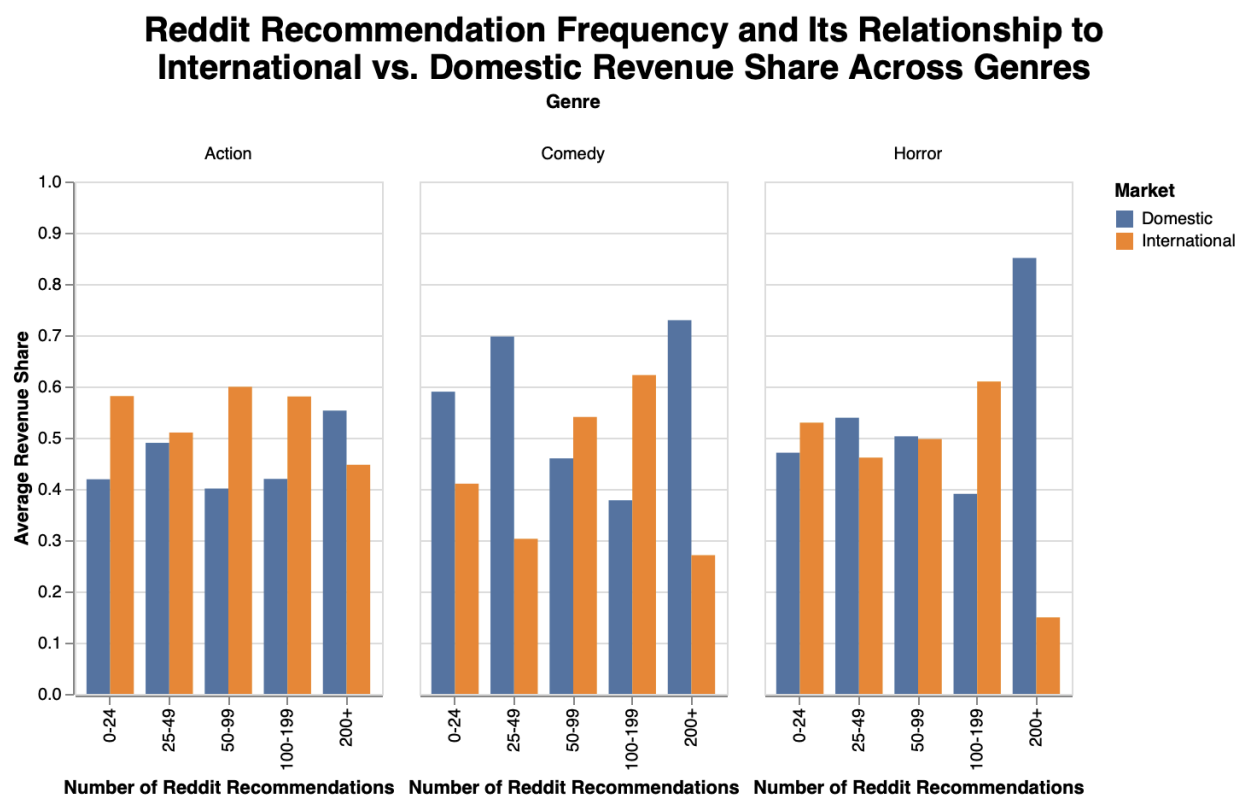
5.

Please refer to “phase4_analysis.ipynb”

This document uses the extra packages Matplotlib, Seaborn, and Altair. Please install these before running the document if not already installed.

6. Results and Visualizations

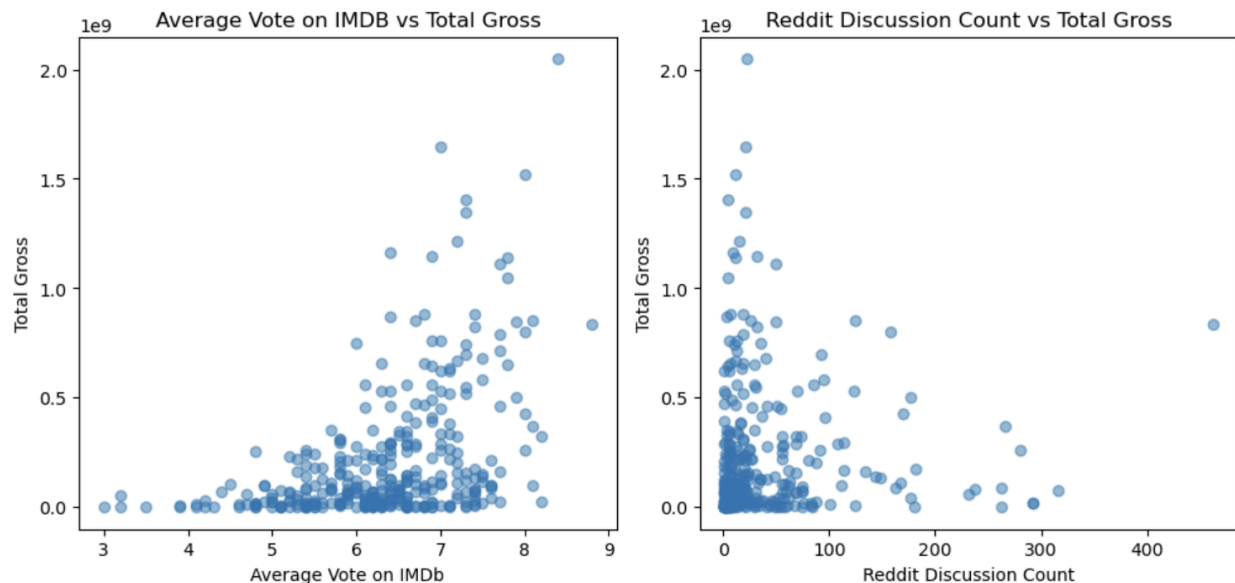
RQ1: How does Reddit recommendation frequency relate to the proportion of a movie’s box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?



This visualization shows how Reddit recommendation frequency relates to the share of box office revenue earned domestically versus internationally across action, comedy, and horror films. For action films, international revenue tends to concentrate from 0-199 recommendations. However, at the highest recommendation level (200+), the domestic share is larger. In contrast, horror films show no clear trend, as domestic and international shares remain relatively balanced across most recommendation levels, although domestic revenue is notably higher at the 200+

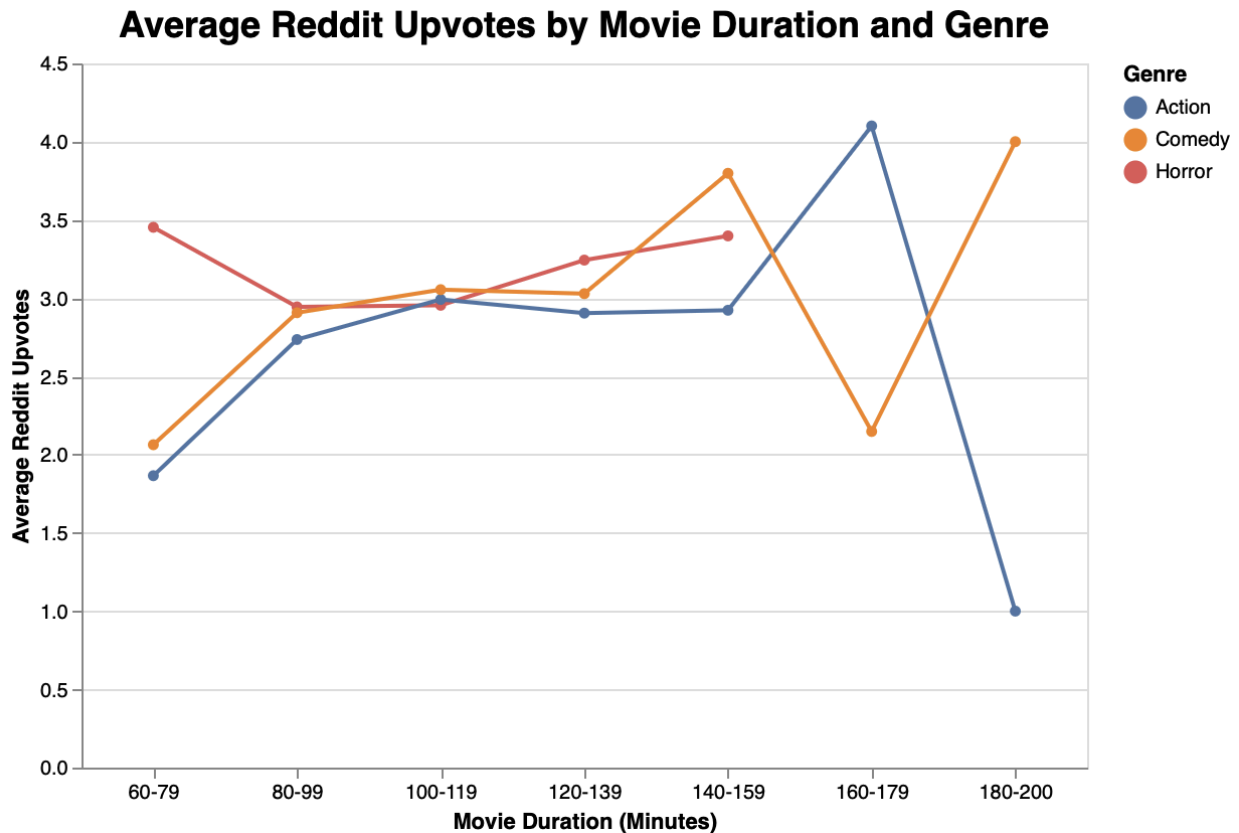
level. For comedy films, domestic revenue consistently exceeds international revenue at most recommendation levels. Overall, Reddit recommendation frequency is not uniformly associated with revenue distribution; comedy and horror films have stronger domestic performance, while action films generally maintain greater international appeal, except at the highest recommendation level.

RQ2: How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?



The visualization above displays a small multiples scatter plot showing total gross vs average vote on IMDb and Reddit discussion count. Scatter plots were chosen to gain visual insight into these relationships prior to conducting regression analyses. From the right plot, it can be inferred that Reddit discussion count is not strongly related to Total Gross. However, the left plot indicates there may be a positive relationship between Average Vote on IMDb and Total Gross. Both graphs show heavy clustering at low Total Gross values, indicating that most movies generate relatively low revenue regardless of IMDb rating or Reddit Discussion Count. This is especially seen in the left plot, which may affect the strength and reliability of a regression model fitted to this data. Both graphs also contain some outliers with very high Total Gross values. This is important to acknowledge in further analysis, as outliers may influence the regression model and should be considered when interpreting the results. Overall, this visualization suggests that Average Vote on IMDb has a more promising relationship to a movie's Total Gross than Reddit Discussion Count, as the right graph displays little association between the variables.

RQ3: How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre (horror, action or comedy)?



This visualization is a line chart that plots how the average number of Reddit upvotes varies across different movie duration bins for action, comedy, and horror films. The lines highlight the trends across duration intervals, while colour is encoded to enable comparison between genres. For action films, upvotes generally increase with duration up to the 160 - 179 minute range, suggesting that longer action films are associated with more Reddit engagement. However, the steep decline in the longest duration bin (180 - 200 minutes) suggests there may be diminishing returns to the average number of Reddit upvotes as the movie duration increases past a certain point. For comedy films, the average number of Reddit upvotes peaks for movies with mid-length durations and very long durations with a drop in between, suggesting a weak relationship between duration and engagement. In horror films, upvotes remain relatively consistent across durations, indicating no clear relationship between movie length and engagement. The lack of data for horror movies in the longer duration categories limits interpretation, as it is unclear whether there is an absence of Reddit mentions for movies that fall into those groups or an absence of movies in those groups entirely. Overall, while longer durations may be associated with higher upvotes in action films, this pattern does not generalize to comedy or horror, indicating that the effect of duration on audience engagement varies by genre.

7. AI Tool Use Declaration

AI Tool: ChatGPT

Prompt: “What aggregations should be used in a mongoDB pipeline to create clean labeled bins for visualizations? Please show me an example using placeholders.”

Response:

Example with more realistic placeholder names:

```
</> JavaScript

db.collection.aggregate([
  {
    $addFields: {
      bin_label: {
        $switch: {
          branches: [
            {
              case: { $lt: ["$VALUE_FIELD", BIN_EDGE_1] },
              then: "BIN_LABEL_1"
            },
            {
              case: { $lt: ["$VALUE_FIELD", BIN_EDGE_2] },
              then: "BIN_LABEL_2"
            },
            {
              case: { $lt: ["$VALUE_FIELD", BIN_EDGE_3] },
              then: "BIN_LABEL_3"
            }
          ],
          default: "BIN_LABEL_4"
        }
      }
    }
  }
])
```

How to read it:

- `$VALUE_FIELD` = the numeric variable you want to bin
- `BIN_EDGE_1`, `BIN_EDGE_2`, ... = cutoff points

AI Tool: Claude

Prompt: For making 2 scatter plots side by side

Response:

```
x = linspace(0,10,50);
y1 = sin(x);
y2 = rand(50,1);
tiledlayout(1,2) // what imports would i need for this code?
```

This looks like **MATLAB** code, not Python — and in MATLAB you don't need any imports. Functions like `linspace`, `sin`, `rand`, and `tiledlayout` are all built-in.

Just run it as-is.

If you're trying to do the **equivalent in Python**, you'd need:

```
python
import numpy as np
import matplotlib.pyplot as plt
```

And the translated code would be:

```
python
x = np.linspace(0, 10, 50)
y1 = np.sin(x)
y2 = np.random.rand(50)

fig, (ax1, ax2) = plt.subplots(1, 2)
```



how to make it less squished?

The titles are overlapping because the two subplots are too close together. A few fixes:

Option 1: Add `plt.tight_layout()` — easiest fix, add it at the end:

```
python
plt.tight_layout()
plt.show()
```

Option 2: Make the figure wider — add this before your subplot code:

```
python
plt.figure(figsize=(14, 6)) # wider figure
```

Option 3: Both together — recommended:

```
python
plt.figure(figsize=(14, 6))
ax1 = plt.subplot(1, 2, 1)
# ... your plot code ...
ax2 = plt.subplot(1, 2, 2)
# ... your plot code ...
plt.tight_layout()
plt.show()
```



AI Tool: Claude

Prompt: “Why isn’t the `reddit_discussion_count > 0` clause working in this (mongoDB):
pipeline2= [

```

{
  "$match": {
    "box_office.domestic_gross": {"$ne": None},
    "box_office.foreign_gross": {"$ne": None},
    "reddit_discussion_count": {"$gt": 0}
  }
},
{
  "$project": {
    "_id": 0,
    "total_gross": {
      "$add": [
        "$box_office.domestic_gross",
        "$box_office.foreign_gross"
      ]
    },
    "avg_vote": 1,
    "reddit_discussion_count": {"$size": "$reddit_mentions"}
  }
},
]”

```

Response:

You just need to add a second `$match` stage after the `$project` to filter on the calculated field:

```

python
pipeline2 = [
  {
    "$match": {
      "box_office.domestic_gross": {"$ne": None},
      "box_office.foreign_gross": {"$ne": None},
    }
  },
  {
    "$project": {
      "_id": 0,
      "total_gross": {
        "$add": [
          "$box_office.domestic_gross",
          "$box_office.foreign_gross"
        ]
      },
      "avg_vote": 1,
      "reddit_discussion_count": {"$size": "$reddit_mentions"}
    }
  },
  {
    "$match": {
      "reddit_discussion_count": {"$gt": 0} # ← filter on the calculated field
    }
  },
]

```